Computational Assessment of Oxygen Availability and Hydrodynamic Shear Stress for Optimized Cell Adhesion in SU-8 Microchambers

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Introduction

Microfluidic platforms provide precise spatial control over cell cultures, but cell viability within gas-impermeable SU-8 chambers depends heavily on a strict balance between biochemical supply and mechanical forces. During initial static cell seeding, cells must remain viable while relying entirely on a finite pool of dissolved oxygen. Once active medium perfusion begins, fluid circulation generates hydrodynamic wall shear stress (WSS) that can cause cell detachment if physiological thresholds are surpassed [1-4]. This study presents a multi-scale computational framework to optimize chamber geometry, ensuring adequate oxygen availability while shielding cells from destructive mechanical stripping.

Methods

A 3D microfluidic chip featuring a 1 mm×1 mm central hexagonal culture chamber was modeled. Fluid flow was governed by the steady-state incompressible Navier-Stokes equations, while transient oxygen transport and cellular consumption were modeled using the convection-diffusion equation with Michaelis-Menten kinetics. Numerical simulations were executed via finite volume analysis in ANSYS Fluent across varying chamber heights (150-300 µm) to map localized WSS and track oxygen depletion timelines over time.

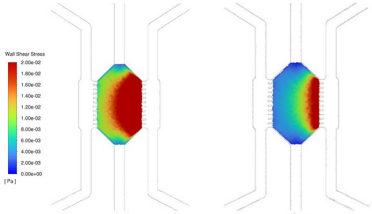
Results

Static simulations revealed that chamber height functions as a critical oxygen reservoir. Based on these numerical results, a novel empirical correlation was derived to predict the temporal evolution of oxygen mass fraction () as a function of chamber height () and time ():

(1)

where is the initial oxygen mass fraction (0.00824), and and are fitting coefficients determined by the cellular metabolic consumption rate (see table 1). The characteristic length scale mathematically reflects the dominant role of diffusion in the sealed chamber.

Under active perfusion (Q=10 µL/min), a restricted 150 µm height induced a hostile mechanical environment exceeding the critical fibroblast detachment threshold (τ_{crit}≈0.02 Pa) [5, 6]. Conversely, doubling the height to 300 µm safely dropped the peak boundary WSS to 0.008 Pa, preventing cell stripping while maintaining necessary convective nutrient delivery [7].



(a) *(b)*

Figure 1: Numerical contours of bottom WSS at Q=10 µL/min for: (a) 150 µm height, and (b) 300 µm height. Red zones highlight areas exceeding the critical cell detachment threshold (τ_{crit}≈0.02 Pa).

Param.	High Consumption	Low Consumption
	7.693	5.409
	2.519	2.535

Table 1: Empirical model parameters for high and low oxygen consumption rates.

Conclusions

The proposed numerical framework successfully delivers clear mechanical guidelines for the design of rigid, thermoplastic organ-on-a-chip platforms. The newly derived empirical correlation offers an accurate mathematical shorthand to predict transient cellular hypoxic risks during initial static cell seeding without requiring computationally intensive CFD simulations. Furthermore, hydrodynamics shows that while the 150 µm height induces destructive fluid shear, the 300 µm design provides a mechanically safe microenvironment, reducing wall shear stress below critical limits to ensure cellular attachment.

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